P2 Chapter 11: Integration

Chapter Practice

Key Points

1 $\int x^n \, dx = \frac{x^{n+1}}{n+1} + c$ $\quad \int e^x \, dx = e^x + c$ $\quad \int \frac{1}{x} \, dx = \ln|x| + c$

$\int \cos x \, dx = \sin x + c$ $\quad \int \sin x \, dx = -\cos x + c$ $\quad \int \sec^2 x = \tan x + c$

$\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$ $\quad \int \operatorname{cosec}^2 x \, dx = -\cot x + c$ $\quad \int \sec x \tan x \, dx = \sec x + c$

2 $\int f'(ax+b) \, dx = \frac{1}{a} f(ax+b) + c$

3 Trigonometric identities can be used to integrate expressions. This allows an expression that cannot be integrated to be replaced by an identical expression that can be integrated.

4 To integrate expressions of the form $\int k \frac{f'(x)}{f(x)} \, dx$, try $\ln|f(x)|$ and differentiate to check, and then adjust any constant.

5 To integrate an expression of the form $\int k f'(x)(f(x))^n \, dx$, try $(f(x))^{n+1}$ and differentiate to check, and then adjust any constant.

6 Sometimes you can simplify an integral by changing the variable. This process is similar to using the chain rule in differentiation and is called **integration by substitution**.

7 The **integration by parts** formula is given by: $\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$

8 Partial fractions can be used to integrate algebraic fractions.

Key Points

9 The area bounded by two curves can be found using integration:

Area of $R = \int_a^b (\mathrm{f}(x) - \mathrm{g}(x))\,\mathrm{d}x = \int_a^b \mathrm{f}(x)\,\mathrm{d}x - \int_a^b \mathrm{g}(x)\,\mathrm{d}x$

10 The **trapezium rule** is:

$$\int_a^b y\,\mathrm{d}x \approx \tfrac{1}{2}h(y_0 + 2(y_1 + y_2 \ldots + y_{n-1}) + y_n)$$

where $h = \dfrac{b-a}{n}$ and $y_i = \mathrm{f}(a + ih)$.

11 When $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\mathrm{g}(y)$ you can write

$$\int \frac{1}{\mathrm{g}(y)}\,\mathrm{d}y = \int \mathrm{f}(x)\,\mathrm{d}x$$

Chapter Exercises

1 By choosing a suitable method of integration, find:

a $\int (2x-3)^7 \, dx$

b $\int x\sqrt{4x-1} \, dx$

c $\int \sin^2 x \cos x \, dx$

d $\int x \ln x \, dx$

e $\int \frac{4 \sin x \cos x}{4 - 8\sin^2 x} \, dx$

f $\int \frac{1}{3-4x} \, dx$

2 By choosing a suitable method, evaluate the following definite integrals. Write your answers as exact values.

a $\int_{-3}^{0} x(x^2+3)^5 \, dx$

b $\int_{0}^{\frac{\pi}{4}} x \sec^2 x \, dx$

c $\int_{1}^{4} \left(16x^{\frac{3}{2}} - \frac{2}{x}\right) dx$

d $\int_{\frac{\pi}{12}}^{\frac{\pi}{3}} (\cos x + \sin x)(\cos x - \sin x) \, dx$

e $\int_{1}^{4} \frac{4}{16x^2+8x-3} \, dx$

f $\int_{0}^{\ln 2} \frac{1}{1+e^x} \, dx$

3 **a** Show that $\int_{1}^{e} \frac{1}{x^2} \ln x \, dx = 1 - \frac{2}{e}$ **(5 marks)**

b Given that $p > 1$, show that $\int_{1}^{p} \frac{1}{(x+1)(2x-1)} \, dx = \frac{1}{3} \ln \frac{4p-2}{p+1}$ **(5 marks)**

4 Given $\int_{\frac{1}{2}}^{b} \left(\frac{2}{x^3} - \frac{1}{x^2}\right) dx = \frac{9}{4}$, find the value of b. **(4 marks)**

5 Given $\int_{0}^{\theta} \cos x \sin^3 x \, dx = \frac{9}{64}$, where $\theta > 0$, find the smallest possible value of θ. **(4 marks)**

Chapter Exercises

6 Using the substitution $t^2 = x + 1$, where $x > -1$,

a find $\int \frac{x}{\sqrt{x+1}}\,dx$. **(5 marks)**

b Hence evaluate $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$. **(2 marks)**

7 **a** Use integration by parts to find $\int x \sin 8x\,dx$. **(4 marks)**

b Use your answer to part **a** to find $\int x^2 \cos 8x\,dx$. **(4 marks)**

8 $f(x) = \frac{5x^2 - 8x + 1}{2x(x-1)^2}$

a Given that $f(x) = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, find the values of the constants A, B and C. **(4 marks)**

b Hence find $\int f(x)\,dx$. **(4 marks)**

c Hence show that $\int_4^9 f(x)\,dx = \ln\left(\frac{32}{3}\right) - \frac{5}{24}$ **(4 marks)**

9 Given that $y = x^{\frac{3}{2}} + \frac{48}{x}$, $x > 0$,

a find the value of x and the value of y when $\frac{dy}{dx} = 0$. **(3 marks)**

b Show that the value of y which you found is a minimum. **(2 marks)**

The finite region R is bounded by the curve with equation $y = x^{\frac{3}{2}} + \frac{48}{x}$, the lines $x = 1$, $x = 4$ and the x-axis.

c Find, by integration, the area of R giving your answer in the form $p + q \ln r$, where the numbers p, q and r are constants to be found. **(4 marks)**

Chapter Exercises

10 **a** Find $\int x^2 \ln 2x \, dx$. **(6 marks)**

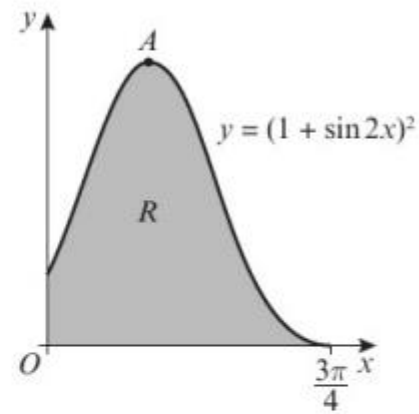
b Hence show that the exact value of $\int_{\frac{1}{2}}^{3} x^2 \ln 2x \, dx$ is $9\ln 6 - \frac{215}{72}$ **(4 marks)**

11 The diagram shows the graph of $y = (1 + \sin 2x)^2$, $0 \leqslant x \leqslant \frac{3\pi}{4}$

a Show that $(1 + \sin 2x)^2 \equiv \frac{1}{2}(3 + 4\sin 2x - \cos 4x)$. **(4 marks)**

b Hence find the area of the shaded region R. **(4 marks)**

c Find the coordinates of A, the turning point on the graph. **(3 marks)**



12 **a** Find $\int x e^{-x} \, dx$. **(4 marks)**

b Given that $y = \frac{\pi}{4}$ at $x = 0$, solve the differential equation

$$e^x \frac{dy}{dx} = \frac{x}{\sin 2y}$$ **(4 marks)**

13 **a** Find $\int x \sin 2x \, dx$. **(5 marks)**

b Given that $y = 0$ at $x = \frac{\pi}{4}$, solve the differential equation $\frac{dy}{dx} = x \sin 2x \cos^2 y$. **(5 marks)**

14 **a** Obtain the general solution to the differential equation

$$\frac{dy}{dx} = xy^2,\ y > 0$$ **(3 marks)**

b Given also that $y = 1$ at $x = 1$, show that

$$y = \frac{2}{3 - x^2},\ -\sqrt{3} < x < \sqrt{3}$$

is a particular solution to the differential equation. **(3 marks)**

The curve C has equation $y = \frac{2}{3 - x^2},\ x \neq \pm\sqrt{3}$

c Write down the gradient of C at the point (1, 1). **(1 mark)**

d Hence write down an equation of the tangent to C at the points (1, 1), and find the coordinates of the point where it again meets the curve. **(4 marks)**

15 **a** Using the substitution $u = 1 + 2x$, or otherwise, find

$$\int \frac{4x}{(1 + 2x)^2}\,dx,\ x \neq -\tfrac{1}{2}$$ **(5 marks)**

b Given that $y = \frac{\pi}{4}$ when $x = 0$, solve the differential equation

$$(1 + 2x)^2 \frac{dy}{dx} = \frac{x}{\sin^2 y}$$ **(5 marks)**

Chapter Exercises

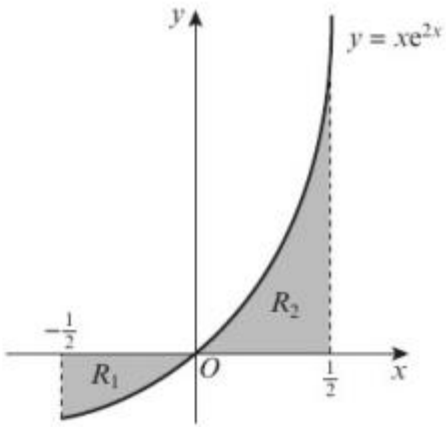
16 The diagram shows the curve with equation $y = xe^{2x}$, $-\frac{1}{2} \leqslant x \leqslant \frac{1}{2}$

The finite region R_1 bounded by the curve, the x-axis and the line $x = -\frac{1}{2}$ has area A_1.

The finite region R_2 bounded by the curve, the x-axis and the line $x = \frac{1}{2}$ has area A_2.

a Find the exact values of A_1 and A_2 by integration. **(6 marks)**

b Show that $A_1 : A_2 = (e - 2) : e$. **(4 marks)**



17 **a** Find $\int x^2 e^{-x}\,dx$. **(5 marks)**

b Use your answer to part **a** to find the solution to the differential equation $\frac{dy}{dx} = x^2 e^{3y-x}$, given that $y = 0$ when $x = 0$. Express your answer in the form $y = f(x)$. **(7 marks)**

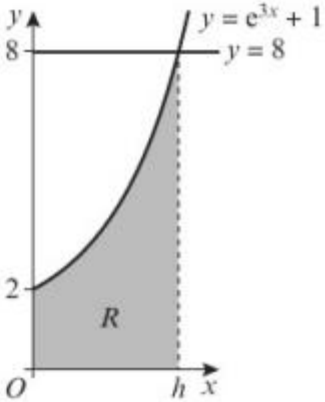
18 The diagram shows part of the curve $y = e^{3x} + 1$ and the line $y = 8$.

The curve and the line intersect at the point $(h, 8)$.

a Find h, giving your answer in terms of natural logarithms. **(3 marks)**

The region R is bounded by the curve, the x-axis, the y-axis and the line $x = h$.

b Use integration to show the area of R is $2 + \frac{1}{3}\ln 7$. **(5 marks)**



Chapter Exercises

19 **a** Given that

$$\frac{x^2}{x^2-1} \equiv A + \frac{B}{x-1} + \frac{C}{x+1}$$

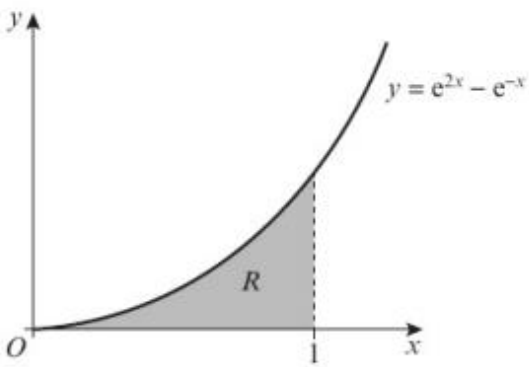
find the values of the constants A, B and C. **(4 marks)**

b Given that $x = 2$ at $t = 1$, solve the differential equation

$$\frac{dx}{dt} = 2 - \frac{2}{x^2}, \; x > 1$$

You do not need to simplify your final answer. **(7 marks)**

20 The curve with equation $y = e^{2x} - e^{-x}$, $0 \leqslant x \leqslant 1$, is shown in the diagram. The finite region enclosed by the curve, the x-axis and the line $x = 1$ is shaded.



The table below shows the corresponding values of x and y with the y values given to 5 decimal places as appropriate.

x	0	0.25	0.5	0.75	1
y	0	0.86992	2.11175		7.02118

a Complete the table with the missing value for y.
Give your answer to 5 decimal places. **(1 mark)**

b Use the trapezium rule, with all the values of y in the table, to obtain an estimate for the area of R, giving your answer to 4 decimal places. **(3 marks)**

c State, with a reason, whether your answer to part **b** is an overestimate or an underestimate. **(1 mark)**

d Use integration to find the exact value of R. Write your answer in the form $\dfrac{e^3 + Pe + Q}{2e}$ where P and Q are constants to be found. **(6 marks)**

e Find the percentage error in the answer to part **b**. **(2 marks)**

Chapter Exercises

21 The rate, in $cm^3 s^{-1}$, at which oil is leaking from an engine sump at any time t seconds is proportional to the volume of oil, $V\,cm^3$, in the sump at that instant. At time $t = 0$, $V = A$.

a By forming and integrating a differential equation, show that

$$V = Ae^{-kt}$$

where k is a positive constant. **(5 marks)**

b Sketch a graph to show the relation between V and t. **(2 marks)**

Given further that $V = \frac{1}{2}A$ at $t = T$,

c show that $kT = \ln 2$. **(3 marks)**

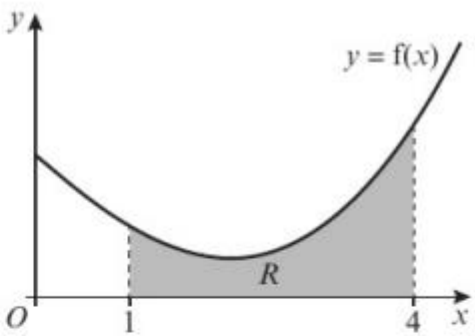
22 **a** Show that the general solution to the differential equation $\frac{dy}{dx} = \frac{x}{k - y}$ can be written in the form $x^2 + (y - k)^2 = c$. **(4 marks)**

b Describe the family of curves that satisfy this differential equation when $k = 2$. **(2 marks)**

23 The diagram shows a sketch of the curve $y = f(x)$, where $f(x) = \frac{1}{5}x^2 \ln x - x + 2$, $x > 0$.

The region R, shown in the diagram, is bounded by the curve, the x-axis and the lines with equations $x = 1$ and $x = 4$.

The table below shows the corresponding values of x and y with the y values given to 4 decimal places as appropriate.



x	1	1.5	2	2.5	3	3.5	4
y	1	0.6825	0.5545	0.6454		1.5693	2.4361

a Complete the table with the missing value of y. **(1 mark)**

b Use the trapezium rule, with all the values of y in the table, to obtain an estimate for the area of R, giving your answer to 3 decimal places. **(3 marks)**

c Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R. **(1 mark)**

d Show that the exact area of R can be written in the form $\frac{a}{b} + \frac{c}{d} \ln e$, where a, b, c, d and e are integers. **(6 marks)**

e Find the percentage error in the answer in part **b**. **(2 marks)**

24 **a** Find $\int x(1 + 2x^2)^5 \, dx$. **(3 marks)**

b Given that $y = \frac{\pi}{8}$ at $x = 0$, solve the differential equation

$$\frac{dy}{dx} = x(1 + 2x^2)^5 \cos^2 2y$$

(5 marks)

Chapter Exercises

25 By using an appropriate trigonometric substitution, find $\int \frac{1}{1+x^2}\,dx$. **(5 marks)**

26 Obtain the solution to

$$x(x+2)\frac{dy}{dx} = y,\ y > 0,\ x > 0$$

for which $y = 2$ at $x = 2$, giving your answer in the form $y^2 = f(x)$. **(7 marks)**

27 An oil spill is modelled as a circular disc with radius r km and area A km^2. The rate of increase of the area of the oil spill, in km^2/day at time t days after it occurs is modelled as:

$$\frac{dA}{dt} = k\sin\left(\frac{t}{3\pi}\right),\ 0 \leqslant t \leqslant 12$$

a Show that $\frac{dr}{dt} = \frac{k}{2\pi r}\sin\left(\frac{t}{3\pi}\right)$ **(2 marks)**

Given that the radius of the spill at time $t = 0$ is 1 km, and the radius of the spill at time $t = \pi^2$ is 2 km:

b find an expression for r^2 in terms of t **(7 marks)**

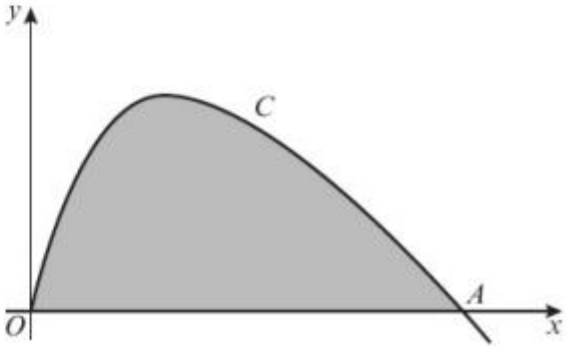
c find the time, in days and hours to the nearest hour, after which the radius of the spill is 1.5 km. **(3 marks)**

28 The diagram shows the curve C with parametric equations

$$x = 3t^2,\ y = \sin 2t,\ t \geqslant 0.$$

a Write down the value of t at the point A where the curve crosses the x-axis. **(1 mark)**

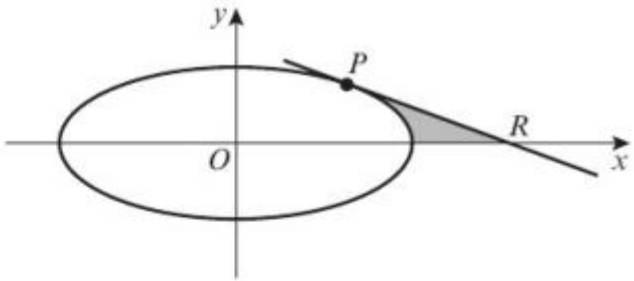
b Find, in terms of π, the exact area of the shaded region bounded by C and the x-axis. **(6 marks)**



Chapter Exercises

29 The curve shown has parametric equations

$$x = 5\cos\theta,\ y = 4\sin\theta,\ 0 \leqslant \theta \leqslant 2\pi$$



a Find the gradient of the curve at the point P at which $\theta = \frac{\pi}{4}$ **(3 marks)**

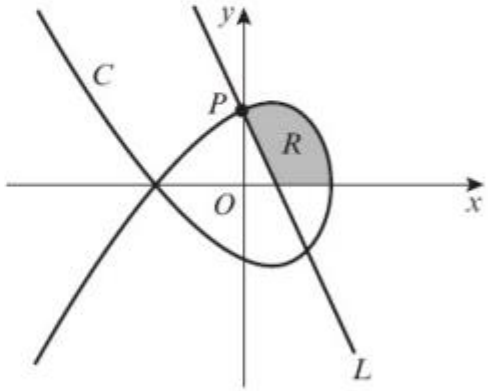
b Find an equation of the tangent to the curve at the point P. **(3 marks)**

c Find the exact area of the shaded region bounded by the tangent PR, the curve and the x-axis. **(6 marks)**

30 The curve C has parametric equations

$$x = 1 - t^2,\ y = 2t - t^3,\ t \in \mathbb{R}$$

The line L is a normal to the curve at the point P where the curve intersects the positive y-axis. Find the exact area of the region R bounded by the curve C, the line L and the x-axis, as shown on the diagram. **(7 marks)**



Challenge

Given $f(x) = x^2 - x - 2$, find:

a $\int_{-3}^{3} |f(x)|\,dx$ **b** $\int_{-3}^{3} f(|x|)\,dx$

Hint Draw a sketch of each function.

Chapter Answers

1 **a** $\frac{1}{16}(2x-3)^8 + c$ **b** $\frac{1}{40}(4x-1)^{\frac{5}{2}} + \frac{1}{24}(4x-1)^{\frac{3}{2}} + c$

c $\frac{1}{3}\sin^3 x + c$ **d** $\frac{x^2}{2}\ln x - \frac{1}{4}x^2 + c$

e $-\frac{1}{4}\ln|\cos 2x| + c$ **f** $-\frac{1}{4}\ln|3-4x| + c$

2 **a** $-\frac{995085}{4}$ **b** $\frac{1}{4}\pi - \frac{1}{2}\ln 2$ **c** $\frac{992}{5} - 2\ln 4$

d $\frac{\sqrt{3}-1}{4}$ **e** $\frac{1}{4}\ln\left(\frac{35}{19}\right)$ **f** $\ln\left(\frac{4}{3}\right)$

3 **a** $\int \frac{1}{x^2}\ln x\,dx = (\ln x)\left(-\frac{1}{x}\right) - \int\left(-\frac{1}{x}\right)\left(\frac{1}{x}\right)dx$

$$= -\frac{\ln x}{x} + \int \frac{1}{x^2}dx = -\frac{\ln x}{x} - \frac{1}{x} + c$$

$$\int_1^e \frac{1}{x^2}\ln x\,dx = \left[-\frac{\ln x}{x} - \frac{1}{x}\right]_1^e = \left(-\frac{1}{e} - \frac{1}{e}\right) - (0-1) = 1 - \frac{2}{e}$$

b $\frac{1}{(x+1)(2x-1)} = \frac{A}{x+1} + \frac{B}{2x-1} \Rightarrow A = -\frac{1}{3}, B = \frac{2}{3}$

$$\int_1^p \frac{1}{(x+1)(2x-1)}dx = \int_1^p\left(-\frac{1}{3(x+1)} + \frac{2}{3(2x-1)}\right)dx$$

$$= \left[-\frac{1}{3}\ln(x+1) + \frac{1}{3}\ln(2x-1)\right]_1^p = \left[\frac{1}{3}\ln\left(\frac{2x-1}{x+1}\right)\right]_1^p$$

$$= \left(\frac{1}{3}\ln\left(\frac{2p-1}{p+1}\right)\right) - \left(\frac{1}{3}\ln\left(\frac{1}{2}\right)\right)$$

$$= \frac{1}{3}\ln\left(\frac{2(2p-1)}{p+1}\right) = \frac{1}{3}\ln\left(\frac{4p-2}{p+1}\right)$$

4 $b = 2$

5 $\theta = \frac{\pi}{3}$

6 **a** $\frac{2}{3}(x-2)\sqrt{x+1} + c$ **b** $\frac{8}{3}$

7 **a** $-\frac{1}{8}x\cos 8x + \frac{1}{64}\sin 8x + c$

b $\frac{1}{8}x^2\sin 8x + \frac{1}{32}x\cos 8x - \frac{1}{256}\sin 8x + c$

8 **a** $A = \frac{1}{2}, B = 2, C = -1$

b $\frac{1}{2}\ln|x| + 2\ln|x-1| + \frac{1}{x-1} + c$

c $\int_4^9 f(x)\,dx = \left[\frac{1}{2}\ln|x| + 2\ln|x-1| + \frac{1}{x-1}\right]_4^9$

$$= \left(\frac{1}{2}\ln 9 + 2\ln 8 + \frac{1}{8}\right) - \left(\frac{1}{2}\ln 4 + 2\ln 3 + \frac{1}{3}\right)$$

$$= \left(\ln 3 + \ln 64 + \frac{1}{8}\right) - \left(\ln 2 + \ln 9 + \frac{1}{3}\right)$$

$$= \ln\left(\frac{3\times 64}{2\times 9}\right) - \frac{5}{24} = \ln\left(\frac{32}{3}\right) - \frac{5}{24}$$

9 **a** $x = 4, y = 20$

b $\frac{d^2y}{dx^2} = \frac{3}{4}x^{-\frac{1}{2}} + \frac{96}{x^3}$

when $x = 4$, $\frac{d^2y}{dx^2} = \frac{15}{8} > 0 \Rightarrow$ minimum

c $\frac{62}{5} + 48\ln 4$; $p = \frac{62}{5}, q = 48, r = 4$

10 **a** $\frac{1}{3}x^3 \ln 2x - \frac{1}{9}x^3 + c$

b $\left[\frac{1}{3}x^3 \ln 2x - \frac{1}{9}x^3\right]_{\frac{1}{2}}^{3} = (9\ln 6 - 3) - \left(0 - \frac{1}{72}\right)$

$= 9\ln 6 - \frac{215}{72}$

11 **a** $(1 + \sin 2x)^2 \equiv 1 + 2\sin 2x + \sin^2 2x$

$\equiv 1 + 2\sin 2x + \frac{1 - \cos 4x}{2} \equiv \frac{3}{2} + 2\sin 2x - \frac{\cos 4x}{2}$

$\equiv \frac{1}{2}(3 + 4\sin 2x - \cos 4x)$

b $\frac{9\pi}{8} + 1$ **c** $\left(\frac{\pi}{4}, 4\right)$

12 **a** $-x\,e^{-x} - e^{-x} + c$ **b** $\cos 2y = 2e^{-x}(x - e^{x} + 1)$

13 **a** $-\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x + c$

b $\tan y = -\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x - \frac{1}{4}$

14 **a** $-\frac{1}{y} = \frac{1}{2}x^2 + c$

b $x = 1$: $-\frac{1}{1} = \frac{1}{2} + c \Rightarrow c = -\frac{3}{2}$

$-\frac{1}{y} = \frac{1}{2}x^2 - \frac{3}{2} \Rightarrow \frac{1}{y} = \frac{1}{2}(3 - x^2) \Rightarrow y = \frac{2}{3 - x^2}$

c 1 **d** $y = x$; $(-2, -2)$

15 **a** $\ln|1 + 2x| + \frac{1}{1 + 2x} + c$

b $2y - \sin 2y = \ln|1 + 2x| + \frac{1}{1 + 2x} + \frac{\pi}{2} - 2$

16 **a** $A_1 = \frac{1}{4} - \frac{1}{2e}$, $A_2 = \frac{1}{4}$

b $A_1 : A_2 \Rightarrow \frac{1}{4} - \frac{1}{2e} : \frac{1}{4} \Rightarrow 1 - \frac{2}{e} : 1 \Rightarrow (e - 2) : e$

17 **a** $-e^{-x}(x^2 + 2x + 2) + c$

b $y = -\frac{1}{3}\ln|3e^{-x}(x^2 + 2x + 2) - 5|$

18 **a** $\frac{1}{3}\ln 7$

b $\int_0^{\frac{1}{3}\ln 7} (e^{3x} + 1)\,dx = \left[\frac{e^{3x}}{3} + x\right]_0^{\frac{1}{3}\ln 7}$

$= \left(\frac{7}{3} + \frac{1}{3}\ln 7\right) - \left(\frac{1}{3} - 0\right) = 2 + \frac{1}{3}\ln 7$

19 **a** $A = 1$, $B = \frac{1}{2}$, $C = -\frac{1}{2}$

b $x + \frac{1}{2}\ln|x - 1| - \frac{1}{2}\ln|x + 1| = 2t - \frac{1}{2}\ln 3$

20 **a** 4.00932 **b** 2.6254

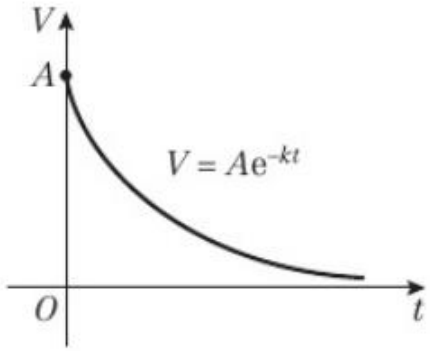
c The curve is convex, so it is an overestimate.

d $\frac{e^3 - 3e + 2}{2e}$, $P = -3$, $Q = 2$ **e** 2.5%

21 **a** $\frac{dV}{dt} = -kV \Rightarrow \int \frac{1}{V}\,dV = \int -k\,dt \Rightarrow \ln V = -kt + c$

$\Rightarrow V = Ae^{-kt}$

b



c $\frac{1}{2}A = Ae^{-kT} \Rightarrow \frac{1}{2} = e^{-kT} \Rightarrow 2 = e^{kT} \Rightarrow \ln 2 = kT$

22 **a** $\int (k - y)\,dy = \int x\,dx \Rightarrow ky - \frac{1}{2}y^2 = \frac{1}{2}x^2 + c$

$x^2 + (y - k)^2 = C$

b Concentric circles with centre (0, 2).

23 **a** 0.9775 **b** 3.074

c Use more values, use smaller intervals. The lines would then more closely follow the curve.

d $\int_1^4 \left(\frac{1}{5}x^2\right) \ln x - x + 2\,dx$

$= \left[\frac{1}{15}x^3 \ln x - \frac{1}{45}x^3 - \frac{1}{2}x^2 + 2x\right]_1^4$

$= \left(\frac{64}{15}\ln 4 - \frac{64}{45}\right) - \left(-\frac{1}{45} - \frac{1}{2} + 2\right) = \frac{-29}{10} + \frac{64}{15}\ln 4$

e 2.0%

24 **a** $\frac{(1 + 2x^2)^6}{24} + c$ **b** $\tan 2y = \frac{1}{12}(1 + 2x^2)^6 + \frac{11}{12}$

25 $\arctan x + c$ **26** $y^2 = \frac{8x}{x + 2}$

27 **a** $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$

$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt} = \frac{1}{2\pi r} \times k \sin\left(\frac{t}{3\pi}\right) = \frac{k}{2\pi r}\sin\left(\frac{t}{3\pi}\right)$

b $r^2 = -6\cos\left(\frac{t}{3\pi}\right) + 7$ **c** 6 days, 5 hours

28 **a** $\frac{\pi}{2}$ **b** $\frac{3\pi}{2}$

29 **a** $-\frac{4}{5}$ **b** $y - 2\sqrt{2} = -\frac{4}{5}\left(x - \frac{5}{\sqrt{2}}\right)$ **c** $10 - \frac{5\pi}{2}$

30 $\frac{41}{60}$

Challenge

a 15 **b** −3